Larger networks and edge electrodes under junction noise

Status report, 13 October 2026. Follows the report of 9 October (ten simulated devices per size; at a realistic junction noise level 9 of 10 devices of 2.83 mm compute beyond their input; 4 mm passed on 3 devices). This report completes 4 mm with 10 devices and compares the edge-electrode layout (split bands) with the pad array on the same tests. Simulations of random Ag nanowire networks at the working density (2000 wires of 70 µm per mm^2), full Kirchhoff circuit with wire resistance (AgNW_FDT_Memristor_Framework), N10 volatile junctions; 690 runs, most on the CCT-Rosario HPC cluster.

> **Legend — color code. green = worked / positive, red = failed / negative, blue = running or to be tested, violet = still to be determined.**
>
> **Jargon in 30 seconds.** A *reservoir computer* uses a fixed physical network as memory and trains only a linear readout. **MC** = memory capacity (past inputs recalled); **degree-2** = nonlinear capacity; **NARMA-10** = a standard task, reported as error (lower is better; input alone 0.84). **G1** = "computes beyond the input": MC ≥ 1, degree-2 ≥ 1, NARMA ≤ 0.70, each with a 2-sd margin over all runs (pooled), or, per device, on the mean of its 2 replicates. A **rupture** is a random, complete break of a junction's volatile filament, at rate r per junction; real junctions are estimated at $r \approx 0.005$–0.03 per second. **Capacity kept** ρ = (MC + degree-2) of a run relative to the same run without ruptures.

Answer

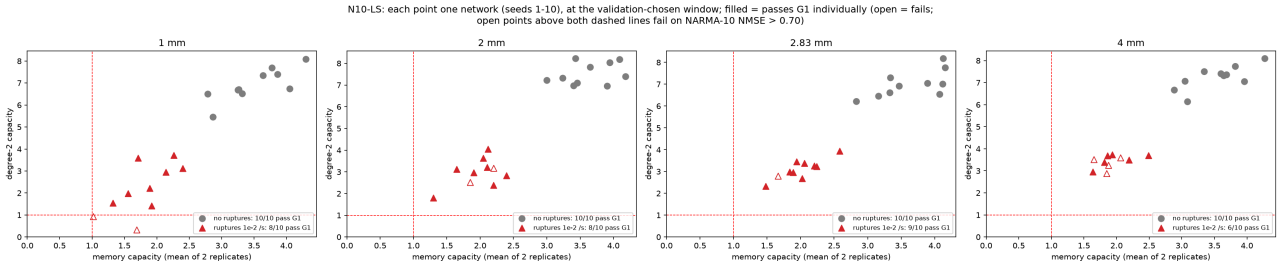
- **4 mm does not confirm.** With 10 devices at 10^{-2} ruptures per second, **6 of 10** 4 mm devices pass G1 and the pooled criterion fails. The 3-device pass of the last report was again luck of the first devices; by the rule fixed in advance the 4 mm claim is withdrawn.
- **The pass rate does not depend on size.** Over 1, 2, 2.83 and 4 mm, 8, 8, 9 and 6 of 10 devices pass (31 of 40). From 2 mm on, memory and nonlinear capacity pass in **every** device; the failures are all on **NARMA-10**, whose error sits just below the 0.70 threshold (0.56-0.66 on average) with a run-to-run spread of about 0.1.
- **Edge electrodes (split bands) give a weaker reservoir than the pad array**: about 30 % less capacity without noise, 15 % less at realistic noise, fewer devices passing G1 (5-7 of 10).
- **But they are more robust to ruptures, and more so the larger the network**: they keep 63 % of their capacity at 2.83 mm (pads 49 %) and almost never collapse (1 of 20 runs, pads 10 of 20).
- **Bias + small input does not help the edge layout** under ruptures (all runs collapse at 10^{-2} per second).

What was run

test	layout	sides	rupture rate (per s per junction)	devices	runs
4 mm completed	pads	4 mm	0 and 10^{-2}	10 (7 new)	84
split bands, size	edge bands	1, 2, 2.83 mm	0 and 10^{-2}	10	408
split bands, rupture ladder	edge bands	1 mm	10^{-4}, 10^{-3}, 3×10^{-3}	3	(in the above)
split bands, bias + small input	edge bands	1 mm	0, 10^{-3}, 10^{-2}	3	162
input window at 1.5 $\times$ onset	both	1 mm	0 and 10^{-2}	3	36

Pads: the 4×4 array of the previous reports (8 driven, 8 grounded, inside the network). Split bands: two edge strips (5 % of the side) cut into 8 driven segments on the left and 8 grounded segments on the right. Readout = the 16 electrode currents in both. Each device: 2 independent streams of 800 inputs at 100 ms per input. **The analysis and the decision rules were written down before the runs** (docs/PLAN_N10L_SEEDS.md, docs/PLAN_SPLIT_BANDS.md).

4 mm with 10 devices



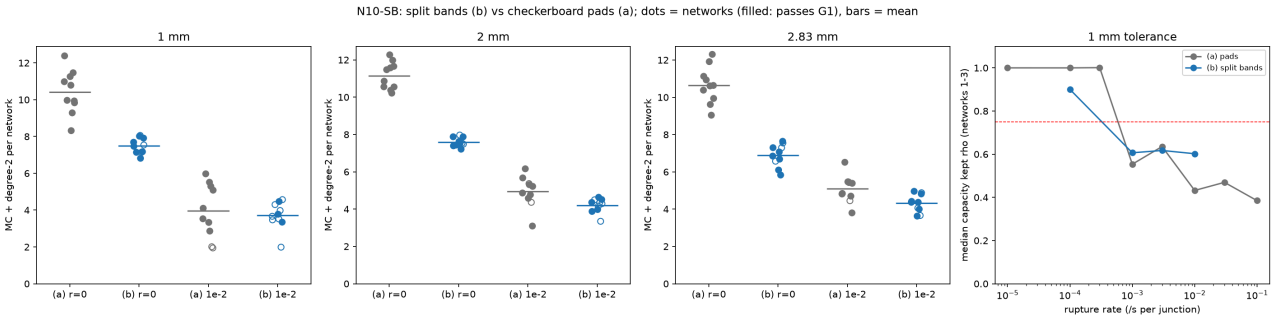
Each point is one device (mean of its 2 runs). Grey: no ruptures; red: 10^{-2} ruptures per second. Filled = passes G1, open = fails. Open points above both dashed lines fail on NARMA-10 only.

side	ruptures /s	memory capacity	degree-2	NARMA-10	pooled G1	devices passing	capacity kept
2.83 mm	10^{-2}	1.99 ± 0.33	3.09 ± 0.55	0.56	no	9/10	0.49

side	ruptures /s	memory capacity	degree-2	NARMA-10	pooled G1	devices passing	capacity kept
4 mm	0	3.54 ± 0.43	7.23 ± 0.53	0.45	yes	10/10	
4 mm	10^{-2}	1.94 ± 0.26	3.41 ± 0.31	0.66	no	6/10	0.51

Mean $\pm$ sd over the 20 runs. The 4 mm devices have the best memory and nonlinear margins of any size and the smallest spread between devices; four of them miss NARMA-10 (0.76-0.90).

Split bands against pads



MC + degree-2 per device (filled = passes G1) for pads (grey) and split bands (blue), without and with 10^{-2} ruptures per second. Right: capacity kept vs rupture rate at 1 mm (3 devices).

side	ruptures /s	MC + degree-2, bands vs pads	devices passing G1, bands / pads	capacity kept, bands / pads	collapsed runs, bands / pads
1 mm	0	7.5 vs 10.4 (worse)	7/10 / 10/10		
1 mm	10^{-2}	3.7 vs 4.0 (on par)	3/10 / 8/10	0.49 / 0.36	12 / 14 of 20
2 mm	10^{-2}	4.2 vs 4.9 (worse)	5/10 / 8/10	0.57 / 0.46	3 / 14 of 20
2.83 mm	0	6.9 vs 10.7 (worse)	7/10 / 10/10		
2.83 mm	10^{-2}	4.3 vs 5.1 (worse)	7/10 / 9/10	0.63 / 0.49	1 / 10 of 20

"Worse" / "on par" / "more robust" by the 95 % bootstrap interval of the difference (rules fixed in advance).

Why: between edge strips the current spreads over many long parallel paths, so a single rupture changes little; between neighbouring pads the paths are few and short and one rupture can reroute much of the current. The same long paths give the edge layout less memory and nonlinearity to start with, and the robustness does not make up for it.

Bias + small input on split bands (the operating point of the measured devices of Milano et al. 2025): without noise it raises capacity by about 40 % (MC 3.2, degree-2 7.2), still well below the biased pads (5.8, 10.9); with ruptures every run collapses, as with the pads.

A side finding. For the pads, an input window at 1.5 $\times$ the switching onset (beyond the two windows used so far) gives more capacity, also with ruptures (3 devices: MC 2.05, degree-2 3.36, NARMA 0.61, all 3 passing). The pad results so far may therefore be slightly conservative. For the split bands it changes nothing.

What this means

- The size claim narrows: **at a realistic junction noise level, about 3 of 4 simulated devices of 1-4 mm compute beyond their input, at about half their noise-free capacity**. Size makes the devices more alike in capacity but does not raise the share that pass; NARMA-10 decides.
- Edge electrodes, the easiest to fabricate, cost capacity; their tolerance to ruptures is better and improves with size, which matters for centimetre devices.
- Ruptures remain the limiting physics for both layouts.

Next

1. Alternating edge layouts (input and output segments interleaved along the edges): edge contacts with a shorter input-output distance, which might combine the robustness of the bands with the capacity of the pads (planned).
2. Re-check the pads at the 1.5 $\times$ onset window with 10 devices (cheap; could restore a pooled pass).
3. Manuscript: literature positioning, journal.
4. Measurements (docs/MEASUREMENT_REQUEST.md).

Names used in this report

name	what it is	outcome
N10-LS	10 devices per size (pads), 1-4 mm	6-9/10 pass at 10^{-2} /s, no size trend

name	what it is	outcome
N10-LS4	the 4 mm part with 10 devices	6/10, pooled fails, claim withdrawn
N10-SB	split bands vs pads (size, rupture ladder, bias, window)	less capacity, more robust from 2 mm
N10-B	bias + small input (pads, earlier)	best noise-free, fragile

Full write-up: docs/RESULTS.md section 27 (N10-LS addendum, update 2026-10-13) and section 28 (N10-SB); plans docs/PLAN_N10L_SEEDS.md, docs/PLAN_SPLIT_BANDS.md; scripts `scripts/n10l_seeds.py`, `scripts/n10sb_layout.py`*; data* `results/n10l_seeds.json`, `results/n10sb.json`, `results/n10sb_bias.json`, `results/n10sb_win15.json`*. Previous reports: docs/report_2026-10-09/, docs/report_2026-10-07/, docs/report_2026-10-06/, docs/report_2026-09-29/.*

The results presented in this work have been obtained by using the facilities of the CCT-Rosario Computational Center, member of the High Performance Computing National System (SNCAD, MincyT-Argentina).